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Losing sight of errors: the effects of typographical conditions on error salience in L2 proof-reading

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Abstract

Proof-reading is the act of focusing on surface error such as grammar, punctuation, and spelling in order to detect deviations from the standard. While much investigation has revealed insights into the correction process, to date we know little of how students go about spotting deviance in their writing, in particular how elements of the reading process itself may impinge on students' ability to do so successfully. This study attempts to examine whether artificially manipulating specific typographical conditions in which text is proof-read promotes better initial recognition of deviant spelling. The study involved 60 Spanish native speakers enrolled in a second-year undergraduate EFL writing class at the University of Granada, all of whom had been previously designated as underachieving. An error recognition task was designed which consisted of four versions of a continuous text in which the same spelling errors were implanted. Statistical analyses of results suggested that subjects' recognition of error was significantly improved by having the text presented in typographical conditions which encouraged concentration on fewer distractors for error detection. Analyses of written and oral protocols revealed that subjects used a number of behaviours which suggested that a conscious effort to counteract the normal process of reading may also help to increase error salience. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Proof-reading has been described as “a subset of revision” and the act of focusing on surface error such as “grammar, punctuation, and...spelling in order to detect deviations from the standard” (Bean and Bouffler, 1987:66). However, while proof-reading is clearly related to reading, different skills are involved in the process. Thus, when we read in the L1, we usually do so for comprehension; our eye movements are rapid, often too rapid to see the details within words — and often failing to see whole words at all (West, 1983). Our knowledge of the language and our general experience help us to identify critical letters or words in a section of text. Readers use their peripheral vision to guide the eye to the most informative part of the line or page. As soon as meaning is grasped, our eyes move on to further units. The brain corrects for omissions and oversights and thereby reduces their importance in the text. It is not surprising, therefore, that simple reading is not an effective proof-reading strategy in the L1 nor in the L2. Ironically, successful proof-reading may depend much more on making a deliberate effort to counteract the normal process of reading.

Advocates of the “process” school of writing have emphasised the need to encourage recursive revision of content throughout the act of writing, claiming that surface proof-reading is typical of “the goals of the novice writer” (Flower, 1981:214). Students writing in strictly ESL contexts may indeed benefit from being encouraged to revise primarily for correctness in content. In a context where opportunities for practice are ubiquitous, instructors can afford to emphasise the priority of precision in content over accuracy in form, confident in the knowledge that surface correctness may be assumed eventually to be acquired by daily use. However, for many less able students writing in EFL academic discourse communities, there are other more pragmatic concerns and needs in writing. Surface correctness may be perceived as a basic requirement for teachers deciding upon a grade (Fitzgerald, 1987), and the ability to spot and rectify deviance quickly may be particularly advantageous in such contexts and even be regarded as manifest proof of written fluency in the L2 (Porte, 1997). In such circumstances, it might also be argued that the skill of proof-reading has a direct effect on these students’ ability to self-monitor and improve their linguistic competence and performance.

What evidence we have seems to suggest that efficient proof-reading consists of two related, yet independent, skills: being able to recognise deviant language and being able to correct that deviance (Zamel, 1983). While much investigation has revealed insights into the correction process itself (Fitzgerald, 1987; Raimes, 1998), to date we know little of how students go about spotting deviance in writing, in particular how elements of the reading process itself may impinge on students’ ability to do so successfully. Ideally, both recognition and correction need to be effected for proof-reading to be beneficial; however, they are not necessarily contiguous, nor are the processes involved as transparent as might first appear. There is evidence that being able to spot a mistake does not lead automatically to its adequate correction by all learners (Gaskill, 1986). Several studies show that poor students are able to sense dissonance in a section of text but may still be unable to pinpoint the

actual problem (Sommers, 1980; Porte, 1997). Once the deviance has been pointed out, however, they tend to be much more successful in bringing about a solution (Raimes, 1991).

The ability to recognise errors may not itself be as straightforward for many poor learners as might first be imagined. Success may depend on more than simply the level of competence reached by the student in that language. A number of psycholinguistic factors have also been shown to affect this recognition process, including script (Englert et al., 1988), L1/L2 interference (Faigley and Witte, 1981), learning strategies (Gaskill, 1986), consciousness-raising, and error fossilisation (Makino, 1993). If such factors do affect the recognition process to any degree, it follows that an inability to correct errors may not, in itself, be a symptom of deficiency in the L2. Furthermore, while general verbal ability in the target language is indeed a prerequisite for proof-reading, there would not appear to be any identifiable skills or specific instruction that guarantee eventual success. Although writers may commit fewer errors as a result of such instruction, their ability to locate deviant language still does not necessarily improve (West, 1983; Lee, 1997).

Certain inherent, and more tangible, characteristics of the reading and writing process may also be hypothesised potentially to impede the salience of L2 written errors for some students. Thus, there are a number of intrinsic elements in the spatial organisation of text on the page such as the shapes and fonts of letters, handwriting, the distances between letters and words, the length of lines, the space between lines, the use of margins, and the use of colour that may all potentially impinge on efficient error recognition.

A small number of studies have investigated the extent to which typographical conditions affect performance in proof-reading, although these have tended to concentrate on L1 writers and their own written production. Nevertheless, a number of findings do support the hypothesis that certain aspects of the reading process affect performance and thus may be susceptible to intervention with the object of facilitating and improving proof-reading. Supramaniam (1983) found that word length and shape had a significant effect on performance on an L1 proof-reading task, indicating that sensitivity to the “gestalt” of a word may be important for successful error recognition. Similarly, Tenney (1980) found that students made less errors in proof-reading for spelling if they were made to write out the possible forms of a word on paper rather than visualise them in their heads. In a study of eye movement patterns in L1 reading texts, Zola (1984) discovered that, in the fixation period, only a certain number of letters can be fixed sharply (between 3 and 5). Thus, word length was hypothesised to influence the time, strategies, and effort involved in proof-reading. A study by Kyle and Wolfe (1990) revealed that two stages are involved in visual processing for revision: (1) an early, broad stage that operates in parallel across the entire visual field; and (2) a later, more circumspect stage, but one which can process information only from a limited area of distractors. Thus, the more distractors in the line of text under scrutiny, the more time and effort the searcher needs. Applying this to L2 proof-reading, if the limits of successful visual processing depend on the amount of distractors to which the reader is exposed, we might hypothesise that by artificially restricting what is exposed for review in each visual

“sweep” of text we may help particularly the less able student to focus on fewer elements and so enhance the salience of certain surface deviance.

There is evidence from intervention studies which supports the hypothesis that such procedural facilitation of the visual perception of text may improve error recognition for less able students. Lydiatt (1984) found that typographical manipulation of between-word spacing aids L1 spelling error correction and hypothesised that raising consciousness by visual means could support other error detection in proof-reading. The study further suggested that there is a fixed span of efficient concentration for comprehension for the English L1 reader of 10 or 11 positions from the centre of vision. Again, it might be postulated that certain L2 surface deviance is less salient for a student because of this mass of text which they intake in one visual sweep, rather than the mere fact that they do not have sufficient knowledge of the target language. It also follows that any instruction in proof-reading would somehow need to intervene in this normal process of reading in order to help the reader focus away from meaning and onto surface form. Madraso (1993) has experimented successfully with “oral reading”, which she reports as forcing her less able L1 students to slow their reading speed (following a metronome), enabling them to spot deviance they missed when reading silently. Porte (1995), in an investigation into the visual perception of L2 text, used computer-generated text margins to “mutilate” text content and showed how consciousness-raising for word images improves grammatical accuracy in poor learners’ writing. He also suggested that such a procedure might facilitate greater attention to other surface features of text. In order to test this hypothesis, this study attempts to examine whether artificially manipulating specific typographical conditions in which text is proof-read promotes better recognition of deviant spelling. Since I was primarily interested in discovering more about the effect of these typographical conditions on visual awareness in initial error recognition, I decided to limit the source of errors only to spelling deviance. My assumption was that initial recognition of such errors depends, arguably more than in other error categories, upon their visual salience for the reader. Hence, the erroneous spelling of an item might be more conspicuous for the poor learner than, say, deviance in word order or subject + predicate agreement.

2. Design of the study

An error recognition task was designed in order to test the hypothesis outlined in the previous section. This consisted of four versions of a continuous text in which the same spelling errors were implanted. These errors were drawn from a database of over 130 examples taken from these same subjects’ written work over the previous 9 months and represented 15 examples of the eight most frequently-encountered spelling error categories in this database:

1. Confusion verb/noun (e.g. “*to develope*”)
2. Single, instead of double, consonants (e.g. “*disatisfaction*”)
3. Double, instead of single, consonants (e.g. “*recommended*”)

4. Short, instead of long, vowels (e.g. “*chosing*”)
5. Letter reversal (e.g. “*enouhg*”)
6. L1 interference (e.g. “*situacion*”)
7. Confusion adverb/noun (e.g. “*dayly*”)
8. Omission of silent letters (e.g. “*enviroment*”)

In order to control for the moderating variables of student script and time spent proof-reading, subjects analysed texts presented on a computer screen in one common font and size and a set time was introduced for the scrutiny of the lines of text made visible (see below). A set time was established in an attempt to simulate the kind of pressure on the error recognition process to which these students are typically exposed in timed examination writing in the faculty, and also thereby discourage the “natural” tendency to concentrate on meaning.

The error recognition task was set in four different typographical conditions:

1. Font: Times New Roman, size 14; double-spacing; average line length — 75 characters.
2. Font: Times New Roman, size 14; double-spacing; average line length — 60 characters.
3. Font: Times New Roman, size 14; double-spacing; average line length — 45 characters.
4. Font: Times New Roman, size 14; double-spacing; average line length — 35 characters.

In each version of the text, different line lengths meant that errors were distributed in a way that ensured their presence throughout the text and in distinct areas of the line. By externally limiting the perceptual span (and, therefore, the number of distractors) available to the student, the assumption was that attention might be more easily fixed on the surface appearance of the text and, therefore, initial error salience enhanced. This meant that subjects would often not be able to form complete sense groups across lines nor concentrate on syntax in the proof-reading. However, the objective was to see how far these writers’ initial recognition of surface deviance can be improved by thus counteracting the normal reading process.

The study seeks to answer the following research questions:

1. Is students’ success rate of correctly detecting spelling error significantly improved by limiting the amount of text visible at any one time?
2. What do oral recordings and written protocols reveal about the recognition process?

3. Subjects

The study involved 60 Spanish native speakers enrolled in a second-year undergraduate EFL writing class at the University of Granada. All the subjects had been previously classified as underachieving in the present course. “Underachievement” was operationalised for the study according to the following criteria: subjects had all

received low band marks (ranging from 60 to 71) in three compositions requiring different discourse types (discursive, narrative and descriptive) from three independent raters using the Jacobs et al. (1981) Composition Profile, had consistently scored less than 40% on average in monthly progress tests of grammar, reading, and listening and were, in the opinion of their current teacher, currently underachieving for their level both in classwork and homework assignments.

Subjects ($n=15$) were randomly assigned to the four conditions of the task. The tasks were administered over a period of 4 weeks towards the end of the academic year. The instructional curriculum had been the same for all subjects and the same teacher had been teaching the writing course. At the time of the study, none of the subjects had received any specific instruction in proof-reading in this course.

4. Procedure

Conditions for the error recognition task were the same across the four versions and the groups. In individual sessions with the researcher, each student sat in front of a computer monitor into which had been typed a short text about a topic previously discussed in class (see Appendix A). The text had earlier been shown to a number of experienced faculty members who all agreed that it was of average level of difficulty for students in the study. A computer-generated frame highlighted four-line sections of the text, simultaneously masking all surrounding lines. Text thus highlighted was displayed for 30 seconds before the researcher moved the frame onto the next four-line section. After each 30-second period, the subject was asked to indicate the errors orally or by pointing to the offending element. Given the aim of the study, only recognition was required and subjects were not asked to correct erroneous items, although many did offer suggestions together with their recognition of errors. To control for possible subject expectancy, students were told that not all lines contained errors but that certain lines might contain more than one error.

The researcher recorded all suggestions for errors on a copy of the particular version of the text being used and also added written comments on performance to provide additional data in the subsequent analysis of protocols. Each task performance was recorded and later transcribed by a research assistant to provide further insights into the recognition process.

5. Results and discussion

5.1. Is students' success rate of correctly detecting spelling error significantly improved by limiting the amount of text visible at any one time?

The error recognition tasks were marked and one point allotted for each correct identification of error. No action was taken in the case of a subject having indicated a correct item as being incorrect, nor in the case of a subject failing to identify a particular error in a line. Thus, the final score (out of 15) for each subject was the

total number of correctly recognised errors. Although standard deviation (S.D.) figures reveal that there was some variability in performance across the four groups, no subject in the sample recorded less than 5 nor more than 11 correct identifications. The results were subjected to statistical analysis using the SPSS package ($\alpha=0.05$). Table 1 shows the mean error recognition scores across the four conditions.

The results show that subjects performed best in Condition 3. Table 1 reveals that the average mean error recognition score was highest in this condition (8.80) and lowest in Condition 1 (6.33). Furthermore, the group variability value recorded (S.D. = 0.82) indicated that students performed most uniformly in Condition 1.

Table 2 presents the results of a one-way ANOVA test on the four means and shows that there was a significant difference among the mean scores of the four conditions ($F=15.293$, $P=0.000$). Although these results confirm considerable differences between results of the four conditions, ANOVA itself fails to identify precisely *where* those differences lie. A post-hoc comparison (Scheffé Test), however, reveals (Table 3) the most significant difference between Conditions 1 and 3 (2.47).

The balanced nature of the research design for this study also allows for a strength of association test (Omega²) to be made, which reveals that 42% of

Table 1
Mean error recognition scores in all conditions

Conditions	Mean	Number	S.D.
1	6.33	15	0.82
2	7.87	15	0.99
3	8.80	15	1.08
4	7.67	15	1.11
Total	7.67	60	1.32

Table 2
Comparison of error recognition scores for all conditions — one-way ANOVA

	df	Sum of squares	Mean squares	F value	P
Between groups	3	46.533	15.511	15.293	0.000
Within groups	56	56.800	1.014		
Total	59	103.333			

Table 3
Post hoc multiple comparison of means differences — Scheffé Test^a

Condition	1	2	3	4
1		1.53*	2.47*	1.33*
2			0.93	0.20
3				1.13*

^a Strength of association test: $\omega^2=0.42$.

* $P<0.05$.

the variability in error recognition scores can be accounted for by the variable “Typographical Condition”.

These findings suggest that subjects’ recognition of error was significantly improved by having the text presented in typographical conditions which encouraged concentration on fewer distractors in text presented for analysis. Furthermore, significant differences were greatest between text of approximately 75 characters per line (Condition 1) and that of 45 characters per line (Condition 3). It is also pertinent to observe that improvement in error recognition only appeared to become consequential when typographical conditions varied considerably. Thus, for example, the improvement seen across means between Conditions 1 and 2 and Conditions 2 and 3 (where the characters per line difference was only 15) was notably less than that observed between Conditions 1 and 3 (30 characters).

Interestingly, however, these differences were not maintained with respect to the condition where least text was visible (Condition 4). Hence, for example, while there were significant differences in means (Table 3) between Condition 3 and Condition 1 (2.47, $P=0.000$), these differences decreased considerably between Condition 4 and Condition 1 (1.33, $P=0.008$) and Condition 4 and Condition 3 (1.13, $P=0.031$) and were not significant at all between Condition 4 and Condition 2 (0.20, $P=0.960$). It would appear that, while overall significant improvement in error recognition was revealed when less text is available for visual sweeping, there may be a point beyond which such typographical conditions — and the consequent “mutilation” of text — no longer enhance the recognition process.

Even allowing for the nature of these subjects (i.e. their confirmed status as underachievers), average scores on this error recognition task were little over 50% of the total available over the four conditions (Total average mean = 7.67/15). Such results may be initially surprising given the fact that the deviance implanted in the task was drawn from the most frequently-encountered errors in these same subjects’ written work and which had also been highlighted and/or corrected concomitantly by the teacher. One can only conjecture as to the reasons for this. However, evidence from L1 studies may provide certain clues. Harris (1987) showed that L1 underachievers will initially read and revise for meaning. Ironically, they may actually be more accustomed to a text which is full of omissions and/or surface imprecision. Using the words and syntax they can recognise and deal with, they attempt to discern meaning as well as they can. It follows then that many such readers will not automatically have their attention drawn to, and will be much less distracted by, these distortions in their texts since meaning will, for them, have been conveyed.

5.2. What do oral recordings and written protocols reveal about the recognition process?

Written comments on student performance recorded certain behaviours which accompanied the error recognition process and which, in turn, may have affected outcomes. Firstly, 21 subjects used a finger to follow the text word-by-word or along the line of text on the monitor, in all but one case halting the finger movement at a doubtful word. As hypothesised at the beginning of this paper, effective

proof-reading may require the reader to counteract the normal process of reading to a certain extent. Oral recordings confirmed that running the finger under or along the text on the monitor usually had the concomitant effect of slowing down the reading, perhaps allowing more time to be spent on deliberation of the current word highlighted. Thus, ironically, what at first may appear a somewhat child-like approach, promoting a linear or serial process of reading may, in fact, be useful as an aid to trigger dissonance (cf. Sommers, 1980).

Secondly, similar evidence for the possible benefits to L2 error salience of counteracting the “normal” reading process were revealed in the oral recordings. Although they were not asked to do so, 18 subjects proceeded with the error recognition task by reading the text aloud or, in the case of 11 others, by mouthing and/or speaking “*sotto voce*” the words seen. Whether such behaviour has any significant effect on success in the error recognition process was not measured directly in this study. However, most recordings did confirm that reading aloud or sub-vocalisation of words would often — in the case of doubtful examples — have the effect of making the subject at least pause and repeat the offending item a number of times in an apparent attempt to focus on, or access, a correct template in the memory, somewhat akin to the habit of writing out a word several times to see which spelling is more likely. One might hypothesise that having the time to enable such verification to be made is indeed central to success in L2 error recognition although not part of the normal L1 or L2 reading process. Thus, checking spelling profits from a more conscious deliberate process than reading and also requires a strong visual memory to handle dissonance. Effective reading is possible, and arguably enhanced, by attending much more selectively to cues in the text. As Madraso (1993) has hypothesised, for the underachiever, it may be that a particular skill is more readily manifested or improved when other cognitive demands are minimised. Thus, the external control here over the amount of text available for analysis — together with the other individually-initiated actions described above — may have combined to decrease or, at least, sufficiently interfere with, the number of otherwise simultaneous reading tasks to be carried out and thereby permit more cognitive space to be allocated to lesser-order tasks such as word-by-word deliberation.

Thirdly, as Sommers (1980) hypothesised, there was evidence that these subjects’ overall underachievement did not necessarily extend to all aspects of their current competence and performance in error detection. While many subjects were unable to detect certain spelling errors, protocols from 38 sessions revealed moments in which these underachievers did indeed indicate or sense something was wrong in the line of text exposed but were unable to pinpoint where exactly the deviance lay. Indeed, 14 subjects were able to point to the general area in the offending line but were unable to proceed, or were wary of doing so.

6. Pedagogical implications

The findings of this study indicate that we might be able to intervene in the process of L2 proof-reading to enhance performance in error recognition. When less

text was made available for analysis, there was significant improvement in error detection. It might be suggested, therefore, that we encourage students to “counteract” the normal process of reading to improve error detection. The idea would be to exploit the natural L1 tendency to fixate on two or three lexical items for meaning but artificially reduce what becomes available in each fixation to enable a more adequate focus on surface deviance. Clearly, as this study has shown, the scrolling and masking features of the computer monitor may help initially in such intervention by dividing up the text into more “manageable” units, but it remains to be seen how far students can be trained to isolate such areas of their own handwritten text.

Results indicate that these underachievers were still not able to spot deviance very successfully despite the fact that the data comprised oft-repeated errors from their own recent writing and to which their attention had been regularly drawn in feedback. As I have suggested, this may be as a result of over-familiarity with the deviance itself as a result of frequent use. As James and Klein (1994) suggest, however, it is possible that errors in words that maintain the main “gestalt” of the target word are more difficult to spot. Even if such aspects are accounted for in the detection process, there still remains the question of what the learner has to search for. If the underachiever has neither the intuition nor the knowledge that there is or might be something wrong with a word, there will remain an indefinite target for the detection process. Hence, one of the crucial factors for successful error detection will be to help students create a kind of sceptical thinking about their own performance. One way in which this may be initially achieved in spelling error detection is by having students practice consciousness-raising using their own errors. One objective here would be to provide a visual image of the part of problem word which locates and highlights the area of difficulty.

Protocol analyses have indicated in this study that detection may be enhanced by activities which, in some way, slow down or otherwise counteract the normal reading process. Oral reading has been shown to achieve this, enabling readers to spot errors they might miss when reading silently. Similarly, as suggested by these findings, when oral reading in a classroom might prove disruptive, a student might try sub-vocalisation together with the use of a finger or pen to point to each word. However, time management would seem to be crucial: subjects here were given only a set time to identify error in an attempt to simulate proof-reading under typical academic writing conditions. As Kroll (1990) indicates, students may find the detection process more successful if some time is allowed to elapse between writing the paper and carrying out the kind of intervention described above.

Finally, there may be lessons from these results for the teacher of writing. These findings call into question just how many marks and symbols, or how much commentary, on students’ surface error may be beneficial and advisable. Given the fact that many of these underachievers were able to sense dissonance if not actually pinpoint it, it might be argued that we should devote our attentions in feedback to encouraging consciousness-raising and thereby enabling students to find their own mistakes since such an activity comes much nearer to the real skill they should be learning. Specifically, such a skill should not involve the abstract understanding of a

mistake someone else has discovered, but rather the detection and correction of errors on own's own.

7. Conclusion

The primary aim of this study was to obtain information about EFL under-achievers' performance in spelling error detection whilst manipulating the variable of typographical conditions. Results have indicated clearly that limited reduction of the text made visible for analysis may be a crucial variable in such detection. Subjects also demonstrated a number of behaviours which indicated that a conscious effort to counteract the normal process of reading may help in highlighting possible errors.

This study used text presented in one font on a computer screen to control for variations in student script. It remains to be seen how far the results obtained in this study are replicated in situations where students are proof-reading their own script. Clearly, in this case, it would be the students themselves — rather than a machine — who must be able eventually to manage how the text is visually presented. Nevertheless, such extra pains during the checking process are clearly worthwhile if students are subsequently better able to spot surface deviance. Furthermore, future research might usefully explore which types of errors — beyond the more visually salient — are enhanced by this kind of heightening of awareness.

Obviously, it cannot be suggested from these findings that simply through the artificial manipulation of typographical and graphological variables we will provide the panacea for poor L2 proof-reading. I do suggest, however, that an important first step towards successful error correction is the ability to sense, and subsequently recognise, errors in one's own production and that attention to the kind of intrinsic graphological and typographical text variables which heighten or lessen student consciousness could prove useful to many at this initial recognition stage.

Appendix A

One of the most important things a university student needs to learn is how to develop good study habits. Some students can study effectively anywhere and any time and on any subject. Many others are dissatisfied because they cannot concentrate on subjects that are difficult or uninteresting to them. Thus, an efficient study area with a chair and table, enough light, and room for study materials is a good place to begin. The time of day and the environment are also important factors. Some students study best in a quiet place early in the morning or late at night whilst others study best with loud music in the middle of the afternoon. Choosing the best time and place to study often depends on a student's family situation and daily schedule. A student who wants to study better has to make some decisions about time, place, and length of study sessions. There are several study methods that can help improve performance. For example, studying the most difficult subject first and

taking a break every hour is often recommended. A brief walk or some stretching exercises can provide a good break. There are also many opportunities to remember more efficiently. Many students use rhymes, mental pictures, and memory aids to help remember information. Although there is no correct way to study, reviewing frequently and carefully following this advice can help organise anyone's study period.

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